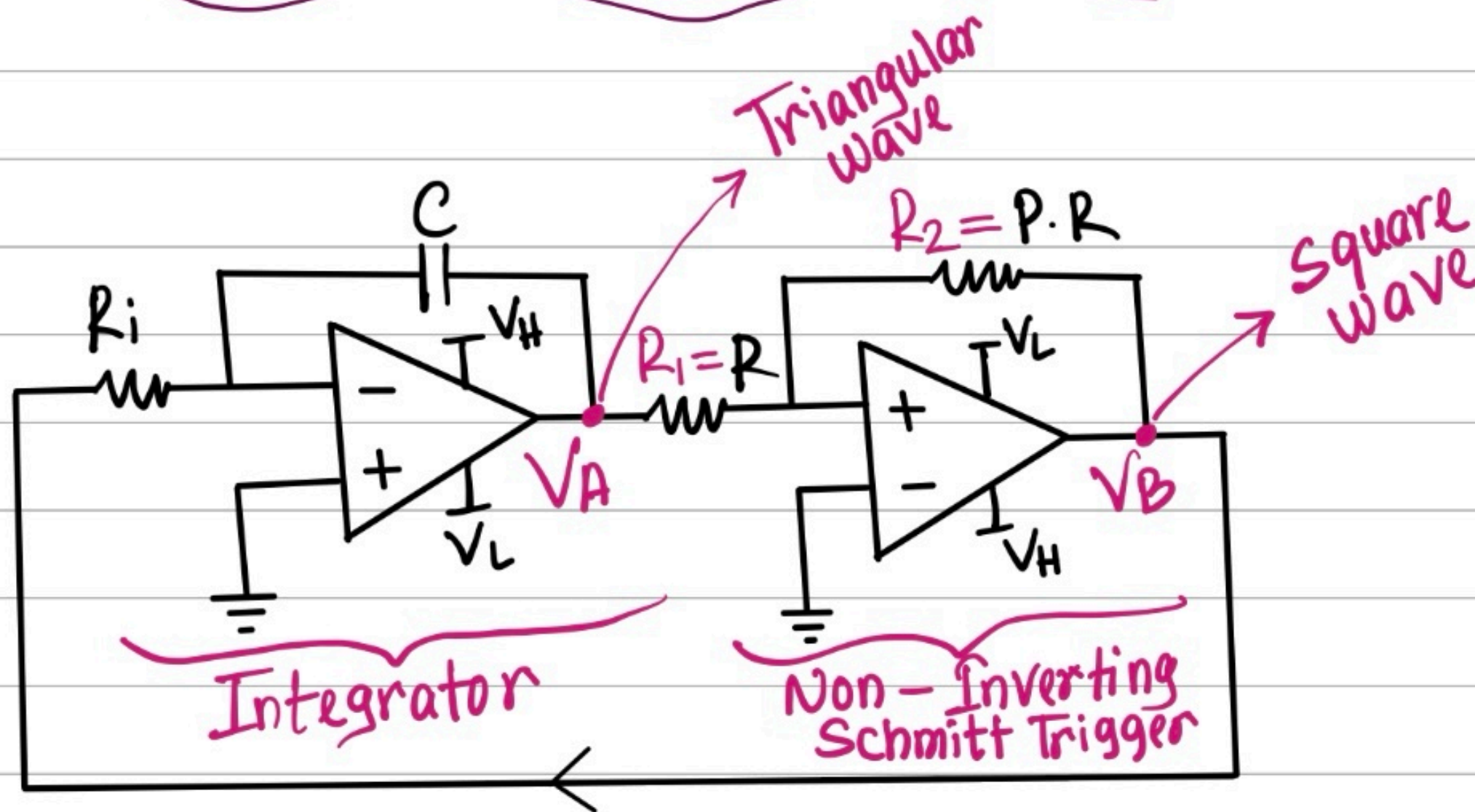


Triangular Wave Generator

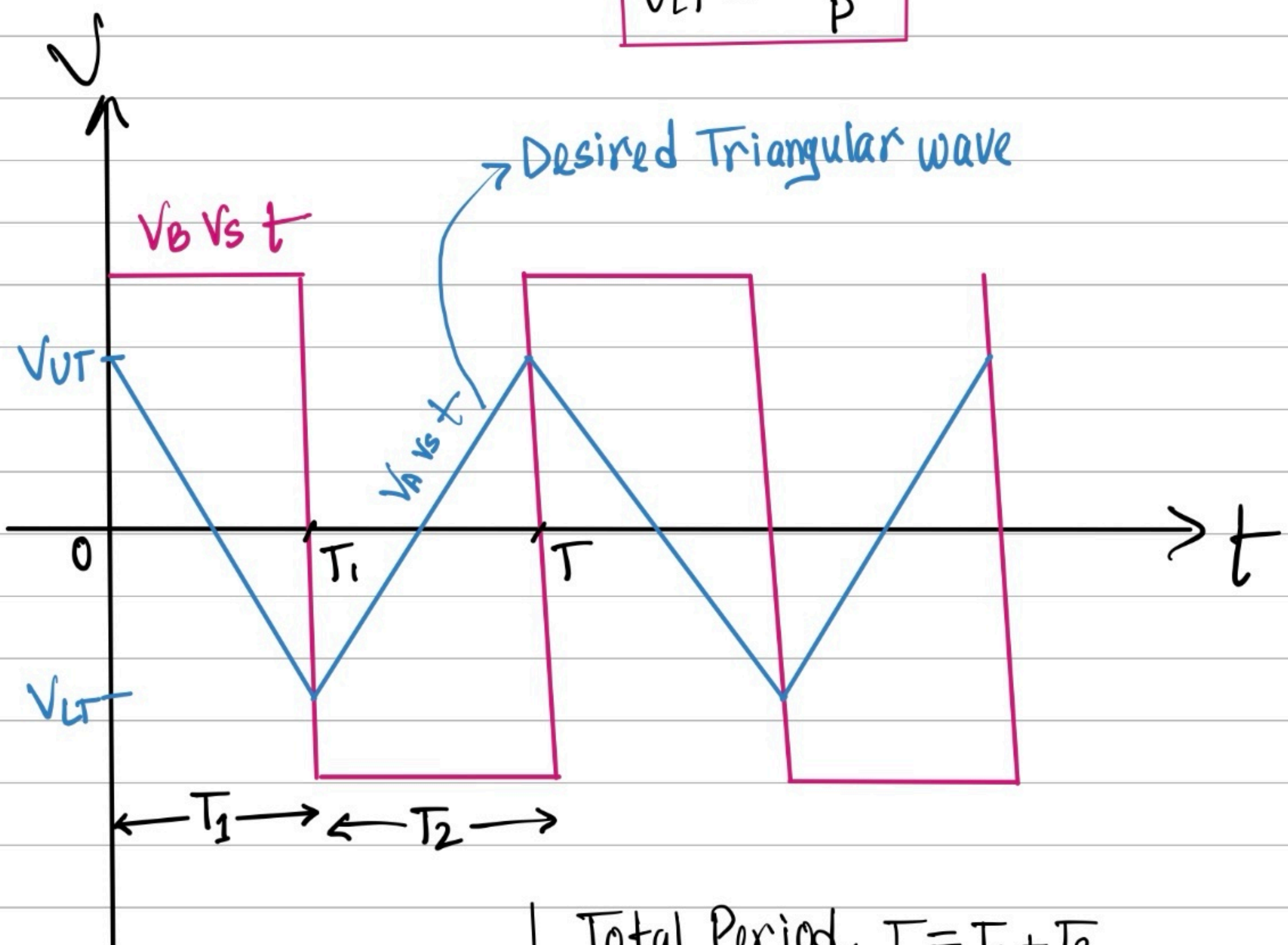


Upper threshold voltage $\rightarrow V_{UT} = -\frac{R_1}{R_2} \times V_L = -\frac{R}{P.R} V_L = -\frac{V_L}{P}$

$$V_{UT} = -\frac{V_L}{P}$$

Lower threshold voltage $\rightarrow V_{LT} = -\frac{R_1}{R_2} \times V_H = -\frac{R}{P.R} V_H = -\frac{V_H}{P}$

$$V_{LT} = -\frac{V_H}{P}$$



$T_1 \rightarrow$ Falling Period
 $T_2 \rightarrow$ Rising Period

Total Period, $T = T_1 + T_2$

Frequency, $f = \frac{1}{T}$

Duty Cycle, $D = \frac{T_2}{T_1 + T_2} \times 100\%$

$$T_1 = R_1 C \left(\frac{V_{UT} - V_{LT}}{V_H} \right)$$

$$T_2 = R_1 C \left(\frac{V_{LT} - V_{UT}}{V_L} \right)$$

** Prove that if $V_L = -V_H \rightarrow$ frequency $f = \frac{P}{4R_1C}$

$\Rightarrow$ given $V_L = -V_H$

$$\begin{cases} V_{UT} = -\frac{V_L}{P} = -\frac{(-V_H)}{P} = \frac{V_H}{P} \\ V_{LT} = \frac{-V_H}{P} \end{cases}$$

$$\therefore T_1 = R_1 C \left(\frac{V_{UT} - V_{LT}}{V_H} \right) = R_1 C \left(\frac{\frac{V_H}{P} - \left(-\frac{V_H}{P}\right)}{V_H} \right) = R_1 C \left(\frac{2V_H}{P V_H} \right) = \frac{2R_1 C}{P}$$

$$\Rightarrow T_1 = \frac{2R_1 C}{P}$$

$$\therefore T_2 = R_1 C \left(\frac{V_{LT} - V_{UT}}{V_L} \right) = R_1 C \left(\frac{-\frac{V_H}{P} - \frac{V_H}{P}}{-V_H} \right) = R_1 C \left(\frac{2V_H}{P V_H} \right) = \frac{2R_1 C}{P}$$

$$\Rightarrow T_2 = \frac{2R_1 C}{P}$$

$$\therefore \text{Period, } T = T_1 + T_2 = \frac{4R_1 C}{P}$$

$$\therefore \text{Frequency, } \left(F = \frac{1}{T} = \frac{P}{4R_1 C} \right) \text{ (Proved)}$$